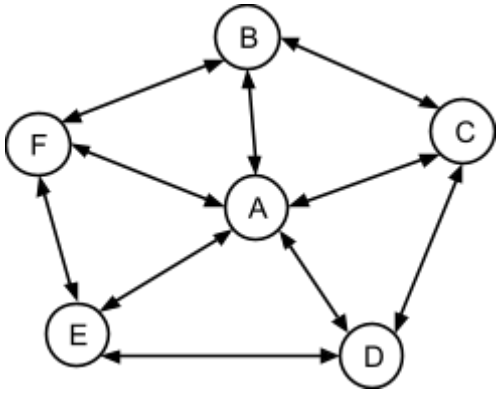
American Computer Science League

2024-2025 ● Contest 4: Short Problems ● Junior Division

1. Graph Theory

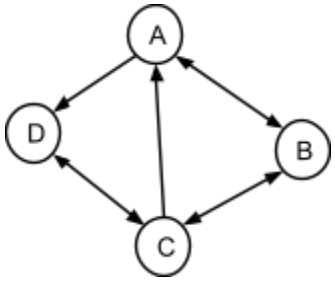
How many 1s are in the adjacency matrix for the following graph?



A. 10
B. 15
C. 20
D. 30
E. 36

2. Graph Theory

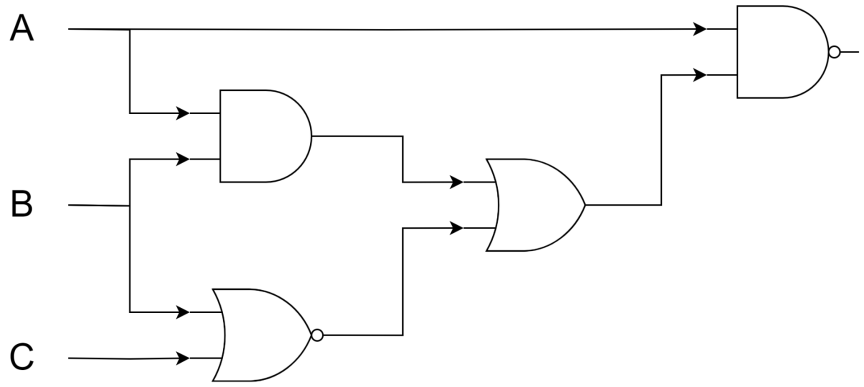
How many cycles are there in the directed graph below?



A. 3
B. 4
C. 5
D. 6
E. 7

3. Digital Electronics

Translate the following digital circuit into a Boolean expression.



A. $\overline{A\,(AB\,+\overline{\overline{B}\,+\,C})}$

B. $\overline{A\,(A\,+\,B\,+\overline{BC})}$

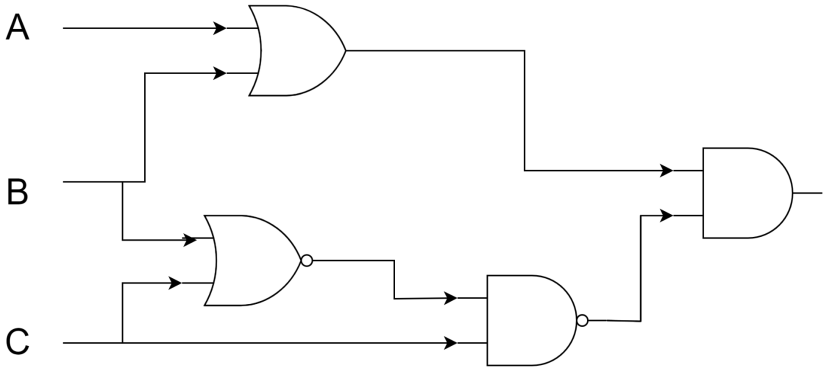
C. $\overline{\overline{A}\,(AB\,+\,\overline{\overline{B}\,+\,C})}$

D. $\overline{A\,+\,AB\,+\overline{BC}}$

E. $\overline{A\,(\overline{AB}\,+\overline{B\,+\,C})}$

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 2
B. 3
C. 4
D. 5
E. 6

5. What Does This Program Do? (Strings)

What will be the output for the following program?

```
s = "JavaPythonC++"
ans1 = s[3:]
ans2 = s[4:9]
ans3 = s[:4]
output ans2 + ans1 + ans3
```

A. `Pytho++JavaP`
B. `PythoC++Jav`
C. `PythonJavaC++`
D. `C++PythonJava`
E. `PythonC++Java`

6. What Does This Program Do? (Strings)

What will be the output for the following program?

```
s = "summeriscoming"
ans = ""
for x = 0 to len(s)/2 - 1
     y = len(s) - x - 1
     if s[x] < s[y] then
          ans = ans + s[x]
     else
          ans = ans + s[y]
     end if
next x
output ans
```

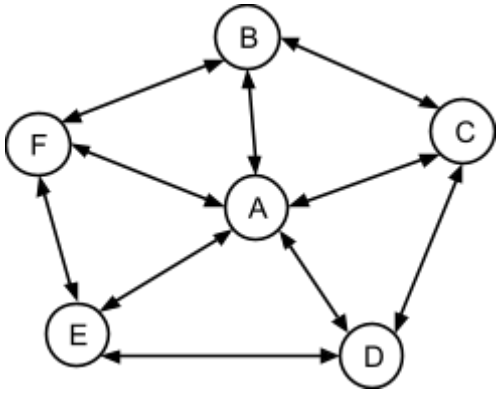
A. `gnimeci`
B. `gnimec`
C. `summors`
D. `gnieci`
E. `gnimocsiremmus`

American Computer Science League

2024-2025 ● Contest 4: Short Problems ● Junior Division

1. 图论

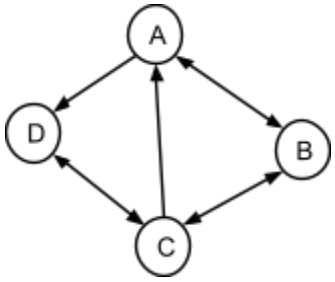
请问下图的邻接矩阵中有多少个 1 ？



A. 10
B. 15
C. 20
D. 30
E. 36

2. 图论

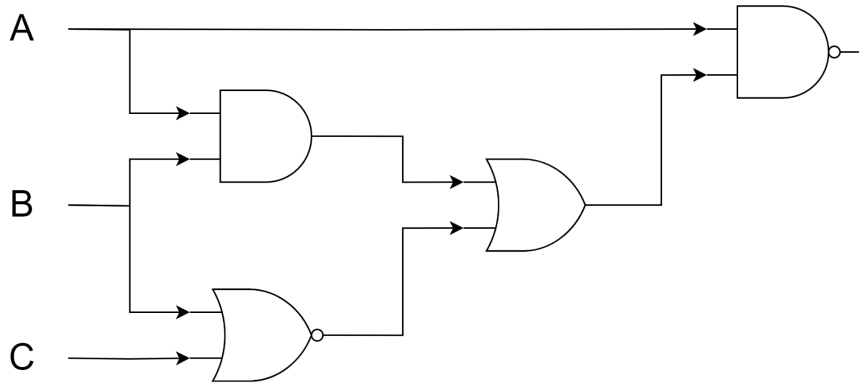
请问以下有向图中存在多少个环？



A. 3
B. 4
C. 5
D. 6
E. 7

3.数字电路

请将以下数字电路转换为布尔表达式。



A. $\overline{A\,(AB\,+\overline{B\,+\,C})}$

B. $\overline{A\,(A\,+\,B\,+\overline{BC})}$

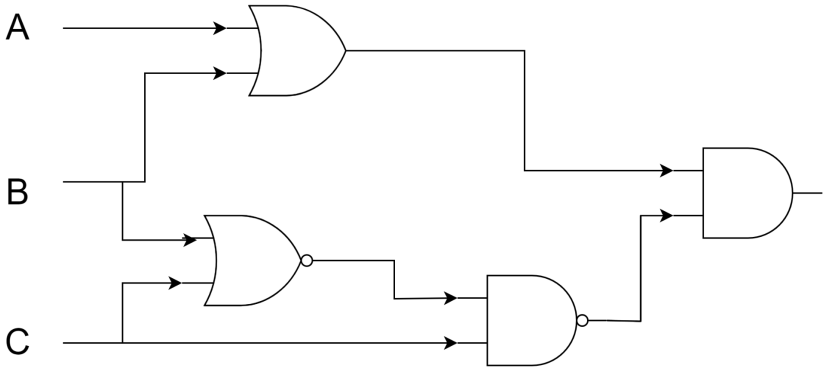
C. $\overline{\overline{A}\,(AB\,+\,\overline{B\,+\,C})}$

D. $\overline{A\,+\,AB\,+\overline{BC}}$

E. $\overline{A\,(\overline{AB}\,+\overline{B\,+\,C})}$

4.数字电路

请问有多少个有序三元组能够使得以下数字电路为真？



A. 2
B. 3
C. 4
D. 5
E. 6

5. 这个程序是做什么的? (字符串)

请问**以下程序的输出结果是什么？**

```
s = "JavaPythonC++"
ans1 = s[3:]
ans2 = s[4:9]
ans3 = s[:4]
output ans2 + ans1 + ans3
```

A. `Pytho++JavaP`
B. `PythoC++Jav`
C. `PythonJavaC++`
D. `C++PythonJava`
E. `PythonC++Java`

6. 这个程序是做什么的? (字符串)

请问**以下程序的输出结果是什么？**

```
s = "summeriscoming"
ans = ""
for x = 0 to len(s)/2 - 1
     y = len(s) - x - 1
     if s[x] < s[y] then
          ans = ans + s[x]
     else
          ans = ans + s[y]
     end if
next x
output ans
```

A. gnimeci
B. gnimec
C. summors
D. gnieci
E. gnimocsiremmus